

★ WEEK 3 · PART 1 of 3

AR, MA, and Wold

The Building Blocks of Time-Series Memory

Lag operator · $AR(p)$ · $MA(q)$ · stationarity & invertibility · ACF/PACF

Once a series is stationary (Week 2), how do we describe its memory? Part 1 introduces the two engines — autoregression and moving averages — and the fingerprints that tell them apart.

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PART 1 · AR & MA

Two Engines of Memory

Autoregression, moving averages, the stationarity and invertibility conditions, Wold's theorem, and the ACF/PACF signatures that identify each.

The Friendship That Wrote the Rulebook — Part 1

How two statisticians gave time series a common language

George Box and Gwilym Jenkins turned a scattered collection of tricks into one disciplined method — a shared grammar for describing and forecasting any stationary series.

The two people

- George Box: a chemist-turned-statistician, famously practical.
- Gwilym Jenkins: a systems and control engineer.
- Together they bridged theory and real forecasting.

The problem they faced

- Practitioners had many ad-hoc models but no common language.
- Choosing among them was an art, not a procedure.
- They wanted a repeatable, teachable method.

The Friendship That Wrote the Rulebook — Part 2

The 1970 book, and the lesson for us

Their 1970 book gave us the ARMA family and the Box-Jenkins method — and Box's enduring reminder: "All models are wrong, but some are useful."

What they delivered

- A unified AR-MA-ARMA framework for stationary series.
- A disciplined loop: identify, estimate, diagnose, repeat.
- It remains the backbone of classical time-series analysis.

The lesson

- A model is a useful approximation, never the truth.
- Judge it by its residuals and its forecasts, not its elegance.
- This humility is the heart of the whole method.

The Big Idea: Describe a Series by Its Memory

Three ways the recent past can shape today

1

Today can echo its own past values

If high values tend to follow high values, the series remembers its own level. That memory is autoregression (AR).

2

Today can echo recent shocks

If a surprise today lingers for a few periods before fading, the series remembers shocks. That memory is a moving average (MA).

3

A series can do both at once

Combine the two and you get ARMA — a compact grammar flexible enough to describe almost any stationary series.

ARMA in one line: explain today using the recent past — its own values and its recent shocks.

White Noise: The Raw Material

Every ARMA model is white noise, filtered

White noise is a stream of independent, zero-mean shocks with no memory. AR and MA models are simply different ways of filtering this raw randomness into structure.

What it is

- Mean zero, constant variance, zero autocorrelation.
- The purest stationary process — no structure at all.
- Denoted ε_t ; the innovation or 'shock' each period.

Why it anchors everything

- AR and MA both take white noise as their input.
- A good model leaves white-noise residuals behind.
- So 'are the residuals white noise?' is the final test.

The Lag Operator

A compact language for the whole ARMA family

$$L X_t = X_{t-1}, \quad \phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$$

What it does

- L shifts a series back one step: $L X_t = X_{t-1}$.
- Powers stack: $L^2 X_t = X_{t-2}$, and so on.
- It turns lag structure into simple polynomials in L .

Why we bother

- AR and MA become polynomials $\phi(L)$ and $\theta(L)$.
- Stationarity and invertibility become root conditions.
- It makes the algebra of ARMA clean and mechanical.

AR(1): The Simplest Memory

Today is a fraction of yesterday, plus a shock

$$X_t = c + \phi X_{t-1} + \varepsilon_t \quad (\text{AR}(1))$$

How to read it

- ϕ is the persistence: how much of yesterday carries over.
- c sets the long-run mean; ε_t is the fresh shock.
- ϕ near 1 means long memory; ϕ near 0 means almost none.

The familiar face

- You met AR(1) in Week 2 as the unit-root borderline.
- Here $\phi < 1$ keeps it stationary and mean-reverting.
- It is the atom from which longer memory is built.

What AR(1) Actually Looks Like

Mean, variance, and autocorrelation in closed form

$$\mathbb{E}[X_t] = \frac{c}{1-\phi}, \quad \text{Var}(X_t) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(k) = \phi^k$$

The three formulas

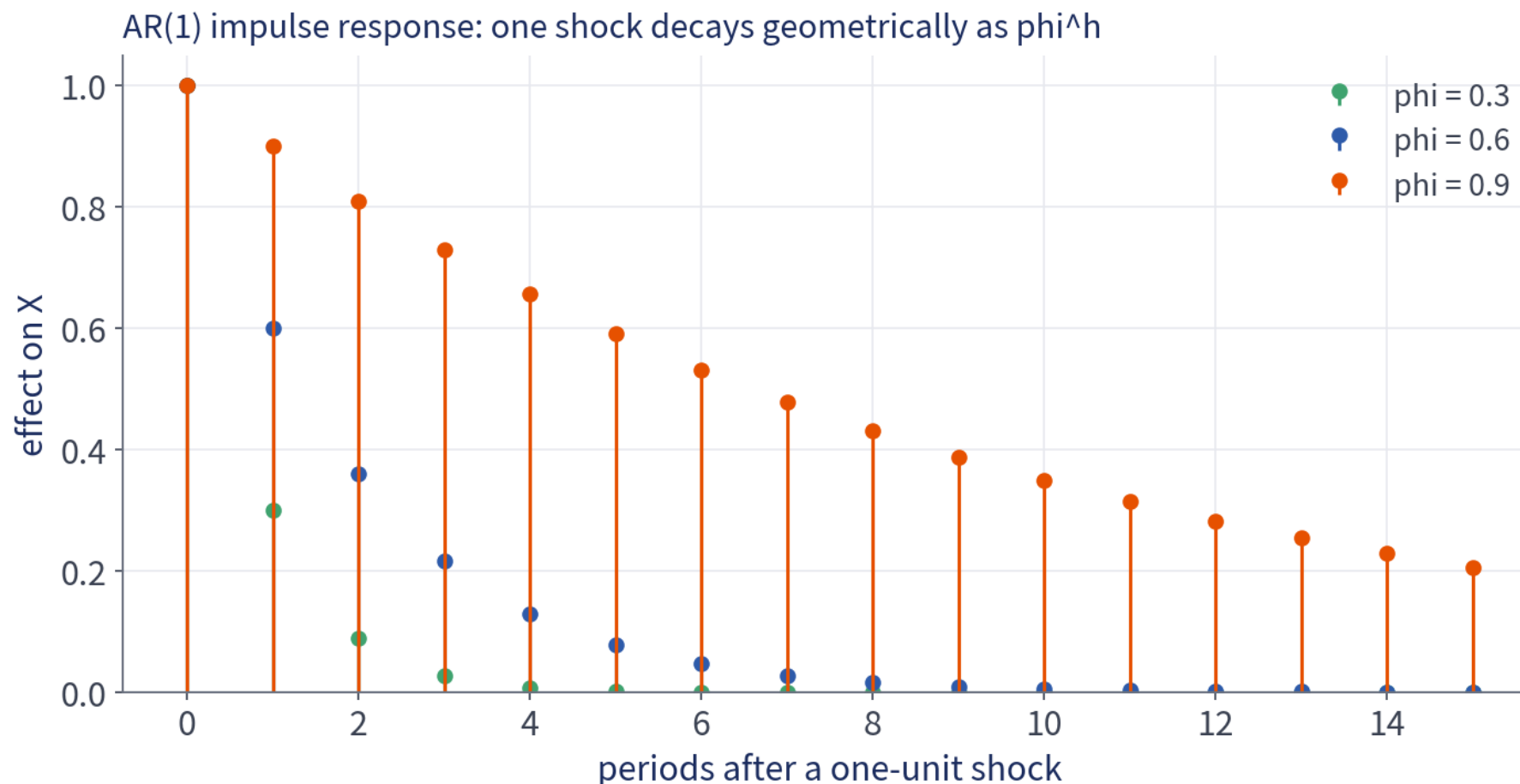
- Mean $c/(1-\phi)$: the level it hovers around.
- Variance grows as ϕ approaches 1 — more persistence, more spread.
- ACF $\rho(k) = \phi^k$: correlation decays geometrically with lag.

The key takeaway

- An AR(1)'s ACF tails off smoothly — it never cuts to zero.
- That smooth decay is the AR fingerprint.
- The next chart shows the same decay as a shock's echo.

How a Single Shock Propagates in an AR(1)

One unit shock today leaves an echo that decays geometrically as ϕ^h — larger ϕ means a longer-lived, slower-fading memory.



AR(p): Longer Memory

Let the last p values all have a say

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t \iff \phi(L)X_t = c + \varepsilon_t$$

The generalization

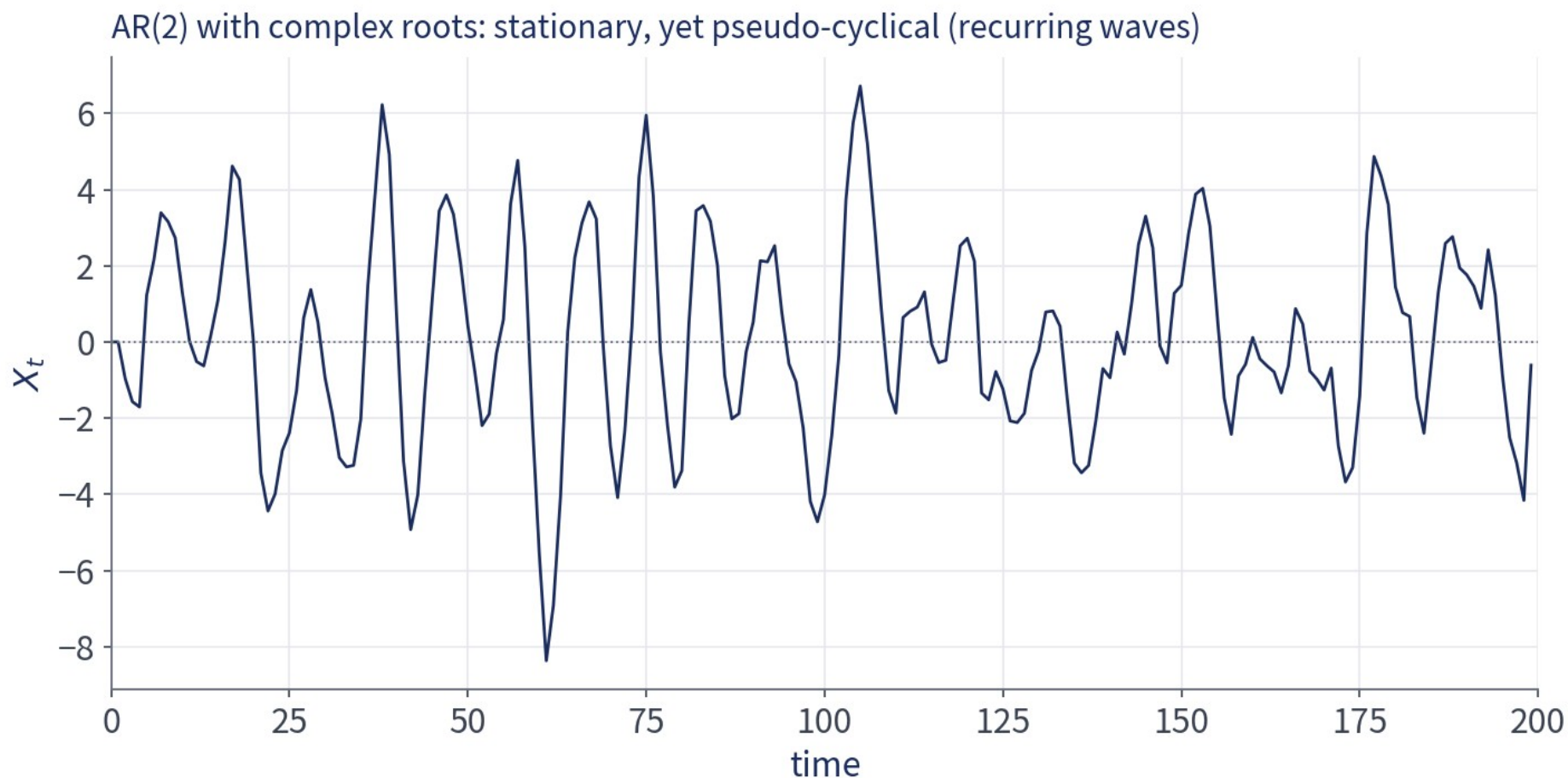
- Regress today on its last p values, each with its own weight.
- In lag-operator form it is simply $\phi(L)X_t = c + \varepsilon_t$.
- p is the order — how far back the direct memory reaches.

What p buys you

- Richer dynamics: overshooting, cycles, slow decay.
- AR(2) can even oscillate (next slide).
- But higher p means more parameters to estimate.

AR(2) Can Be Pseudo-Cyclical

When an AR(2)'s roots are complex, the series is still stationary yet rolls in recurring waves — memory that oscillates rather than simply decays.



When Is an AR Model Stationary?

The roots must lie outside the unit circle

stationary $\iff$ all roots of $\phi(z) = 0$ lie OUTSIDE the unit circle ($|z| > 1$)

The rule

- Write the AR polynomial $\phi(z)$ and find its roots.
- All roots must lie strictly outside the unit circle.
- For AR(1) this reduces exactly to $|\phi| < 1$.

Why it matters

- A root ON the circle is a unit root — non-stationary (Week 2).
- A root INSIDE means an explosive, useless process.
- Every stats package checks this for you automatically.

MA(1): A Finite Echo of Shocks

Surprises matter, but only for a while

$$X_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad \rho(1) = \frac{\theta}{1+\theta^2}, \quad \rho(k) = 0 \quad (k \geq 2)$$

How to read it

- Today is this period's shock plus a fraction of last period's.
- The influence of any shock lasts exactly one lag, then stops.
- θ sets how strongly the previous surprise carries over.

The signature

- Its ACF is non-zero at lag 1, then exactly zero after.
- That sharp cut-off is the MA fingerprint.
- Contrast with AR's smooth, never-ending decay.

MA(q): Echoes of the Last q Shocks

A longer but still finite memory

$$X_t = \mu + \theta(L) \varepsilon_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}$$

The generalization

- Today is a weighted sum of the last q shocks, plus today's.
- In lag form: $X_t = \mu + \theta(L) \varepsilon_t$.
- q is the memory length — how many lags a shock survives.

The clean property

- An MA(q)'s ACF cuts off sharply after lag q.
- So the last non-zero ACF lag reveals q directly.
- This is what makes MA models easy to identify.

Invertibility: The MA Counterpart

So the model has a unique, sensible form

invertible $\iff$ all roots of $\theta(z) = 0$ lie OUTSIDE the unit circle

What it means

- An invertible MA can be rewritten as an infinite AR.
- Its roots, like AR's, must lie outside the unit circle.
- This lets us recover the shocks from the observed data.

Why we require it

- Different θ values can give the SAME autocorrelations.
- Invertibility picks the one, well-behaved representation.
- Without it, estimation and forecasting are ambiguous.

Why MA Is So Fundamental: Wold's Theorem

Every stationary series is an MA at heart

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1 \quad (\text{Wold representation})$$

What it says

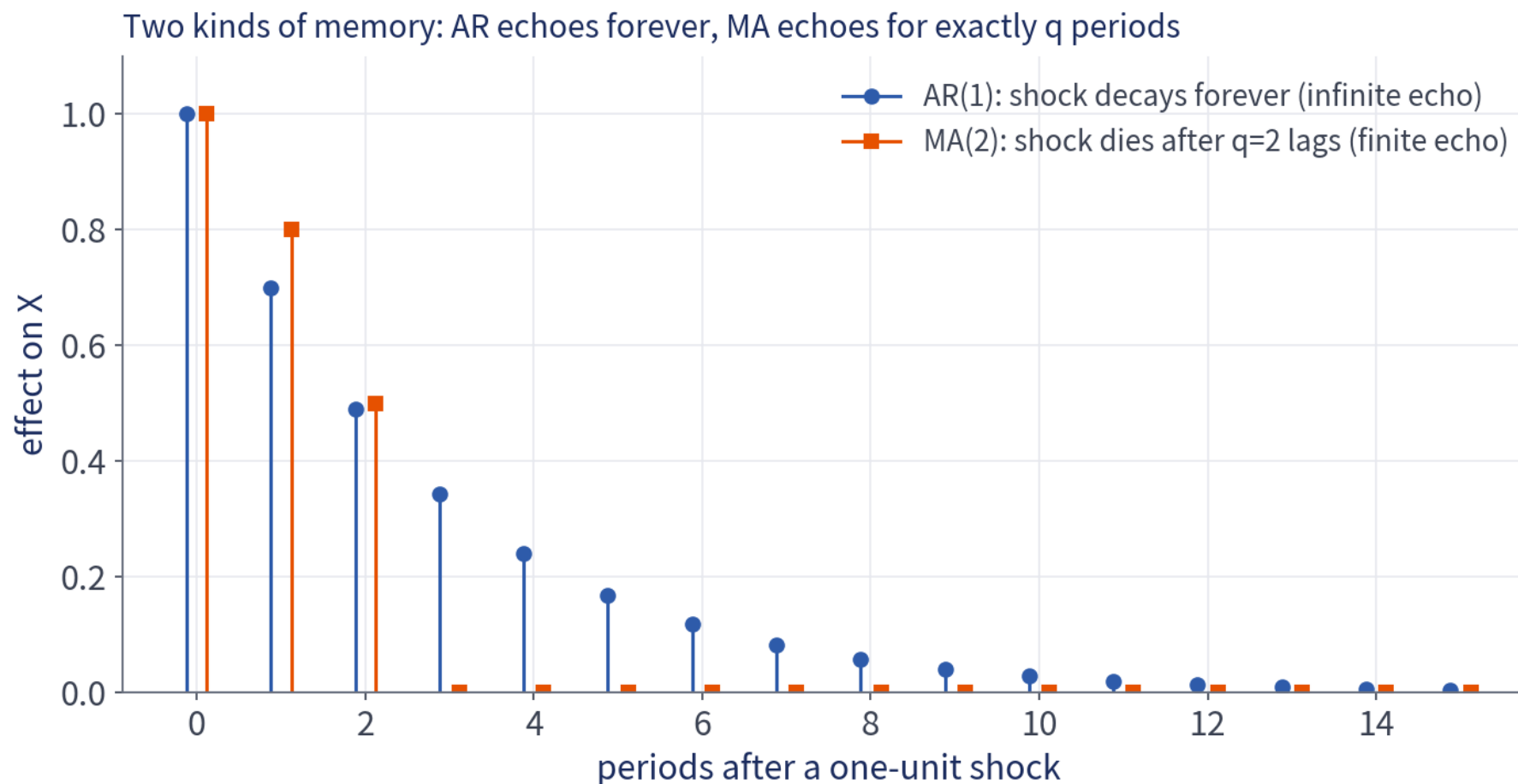
- Any (weakly) stationary series equals an $\text{MA}(\infty)$ of white noise.
- So moving averages are not just one option — they are universal.
- AR and ARMA are just compact ways to write that $\text{MA}(\infty)$.

Why it is reassuring

- It guarantees the ARMA toolbox is broad enough.
- It explains the $\text{AR} \leftrightarrow \text{MA}$ duality we see next.
- It is the theoretical license for the whole method.

AR vs MA: Two Kinds of Memory

Follow a single shock: in an AR it echoes forever, decaying geometrically; in an MA(q) it lives for exactly q periods and then vanishes.



AR and MA Are Two Sides of a Coin

A stationary AR is an infinite MA, and vice versa

$$\underbrace{X_t = \phi X_{t-1} + \varepsilon_t}_{\text{AR}(1)} = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} = \underbrace{\text{MA}(\infty)}_{\text{same process}}$$

The equivalence

- Unrolling an AR(1) turns it into an MA(∞) of past shocks.
- Likewise an invertible MA(1) becomes an AR(∞).
- The two descriptions are the same process, written differently.

Why it is useful

- It lets you pick whichever form is more parsimonious.
- It explains why the ACF and PACF are mirror images.
- It is Wold's theorem made concrete.

The ACF and the PACF

Two views of a series' memory

The ACF measures the TOTAL correlation at each lag; the PACF measures the DIRECT correlation, stripping out what nearer lags already explain. Together they identify a model.

ACF — total correlation

- How correlated is X_t with X_{t-k} , in full?
- For an $MA(q)$ it cuts off sharply after lag q .
- For an AR it tails off smoothly.

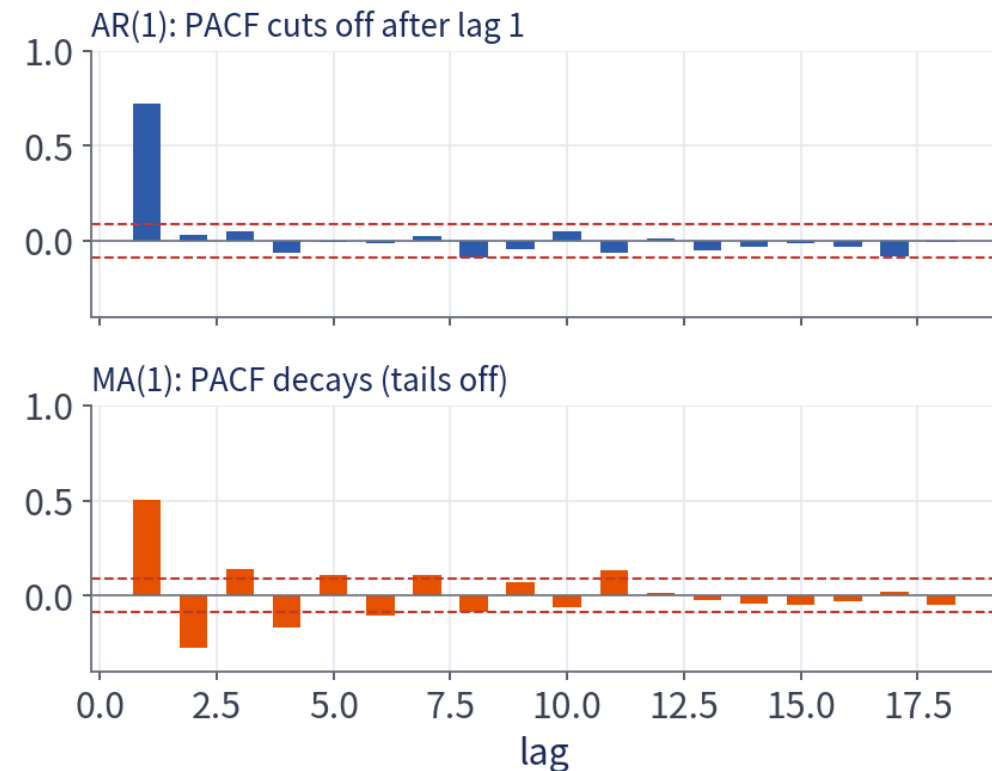
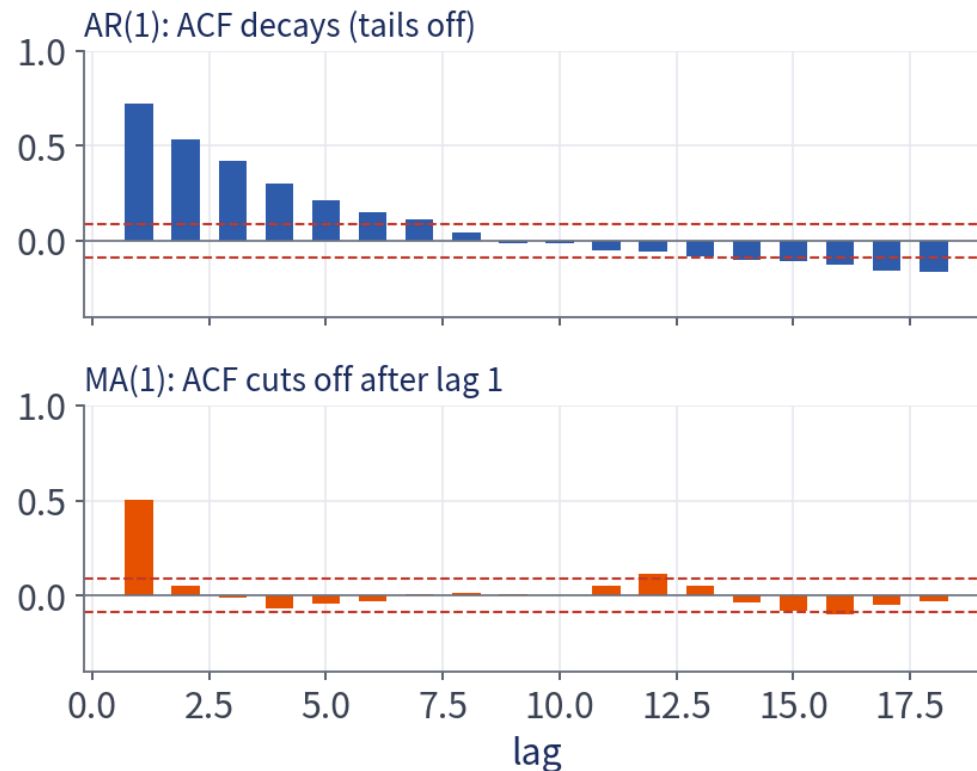
PACF — direct correlation

- The link at lag k after removing lags $1 \dots k-1$.
- For an $AR(p)$ it cuts off sharply after lag p .
- For an MA it tails off smoothly.

ACF / PACF Signatures at a Glance

The single most useful picture in classical time series: whichever plot CUTS OFF tells you the model — ACF cut-off $\rightarrow$ MA(q); PACF cut-off $\rightarrow$ AR(p).

The identification fingerprint: which one cuts off tells you AR vs MA



AR vs MA: Two Engines of Memory

Both turn white-noise shocks into structure, but they remember in opposite ways — and leave opposite fingerprints on the ACF and PACF.

AR vs MA: Two Engines of Memory

Both turn white-noise shocks into structure - but they remember in opposite ways.

AR - Autoregressive (feedback)

Today depends on its own PAST VALUES.

$$X_t = c + \phi X_{t-1} + \dots + e_t$$

One shock feeds back through the level, so its echo decays but never fully dies.

Fingerprint

- ACF: decays (tails off)
- PACF: cuts off after lag p

memory = infinite, geometric

MA - Moving Average (echo)

Today depends on recent SHOCKS.

$$X_t = \mu + e_t + \theta e_{t-1} + \dots$$

A shock echoes for exactly q periods, then vanishes completely.

Fingerprint

- ACF: cuts off after lag q
- PACF: decays (tails off)

memory = finite, exactly q lags

Simulate and Identify

Build the AR/MA fingerprints in your own hands

GOAL Simulate AR and MA processes, then learn to read their ACF/PACF fingerprints before touching real data.

[INPUT] Prompt · Claude Code

Write and run `arma_signatures.py` that:

1. simulates four series of length 800 -
AR(1) $\phi=0.7$, AR(2) with complex roots,
MA(1) $\theta=0.8$, and MA(2);
2. for each, plots the series, its ACF, and
its PACF (with 95% bands) in one row;
3. prints, for each, which plot cuts off and
at what lag - and its implied (p, q).

Then hide the labels and quiz me: from the
ACF/PACF alone, can I name each model?

How to work with it

- AR $\rightarrow$ PACF cuts off; MA $\rightarrow$ ACF cuts off.
- Read the cut-off lag as p or q.
- Re-run with new seeds — do the signatures hold?
- Keep it; Part 2 uses these to identify real data.

Deliverable: `arma_signatures.py` + a one-row ACF/PACF panel per process.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1	A series' ACF cuts off after lag 2 but its PACF tails off. AR or MA, what order?	MA(2)
Q2	Why can every stationary series be written as a moving average?	Wold's theorem
Q3	An AR(1) has $\phi = 1.2$. What is wrong with it?	Explosive (root inside)

All three come straight from the AR/MA fingerprints and the root conditions.

Three Misconceptions to Avoid

Where the fingerprints get misread

Common myths

- "A slowly-decaying ACF means a high MA order."
- "AR and MA are fundamentally different processes."
- "Bigger p and q always fit better."

The reality

- Slow ACF decay points to AR, not MA — read the PACF to confirm.
- By duality they are two descriptions of one process (Wold).
- More parameters overfit; parsimony wins (Part 3's AIC/BIC).

Three Things to Carry Forward

Part 1 in one screen

1 Two engines: AR remembers levels, MA remembers shocks

AR feeds back through past values; MA echoes recent innovations. Stationarity (AR) and invertibility (MA) both come down to roots outside the unit circle.

2 Wold ties it together

Every stationary series is an $MA(\infty)$, and $AR \leftrightarrow MA$ duality means the two forms describe the same process — pick the compact one.

3 The ACF and PACF are your fingerprints

ACF cut-off $\rightarrow MA(q)$; PACF cut-off $\rightarrow AR(p)$. That single reading is the key to identification in Part 2.

Next — Part 2: combining both into ARMA, and the Box-Jenkins method for identifying p and q .